

15 A saturated solution of Ca(OH)_2 is found to have a pH of 12.4 at 25 °C.

[Turn over



Which statements are correct?

- 1 The K_{sp} of $\text{Ca}(\text{OH})_2$ is $7.92 \times 10^{-6} \text{ mol}^3 \text{ dm}^{-9}$.
- 2 The solubility of $\text{Ca}(\text{OH})_2$ will increase when aqueous HCl is added.
- 3 The solubility of $\text{Ca}(\text{OH})_2$ would increase when temperature is raised to 40°C .
- 4 The pH of the solution would increase when solid $\text{Ca}(\text{NO}_3)_2$ is added.

A 1 and 2 only **B** 2 and 3 only **C** 3 and 4 only **D** 1, 2 and 3 only